

Trevor Werbin - Mechanical Engineering Portfolio

I'm a Mechanical Engineering student at the University of Waterloo with hands-on experience in mechanical design, fabrication, and product development. Through my internship at Ciena and course projects, I've developed experience with the iterative design process, parametric CAD, 3D printing, machining, applying DFM principles, GD&T, technical drawings, and product testing. The projects in this portfolio showcase my recent engineering work and how I approach challenges from concept through execution.

Portfolio Projects

1. Autonomous Cookie Icing Machine

Autonomously applies custom icing designs to cookies in <4mins

2. Mechanical Thumb Toy

Wearable, actuating, thumb meeting 30N grip force spec with <1s actuation

3. Solar Car Canopy

Custom canopy modelled in SolidWorks for the University of Waterloo Midnight Sun solar car optimizing for aerodynamics and driver visibility

4. DIY Dynamometer

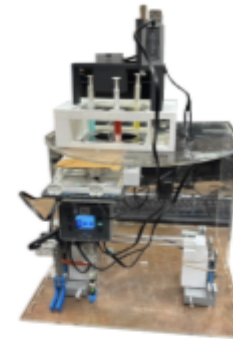
Rope brake dynamometer built to measure torque output of a TT motor achieving results within 45% of manufacturer's specifications

5. Spur Gear Keychain

Functional keychain designed in SolidWorks and machined from aluminum stock

Contact Information

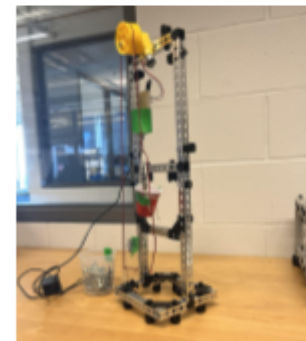
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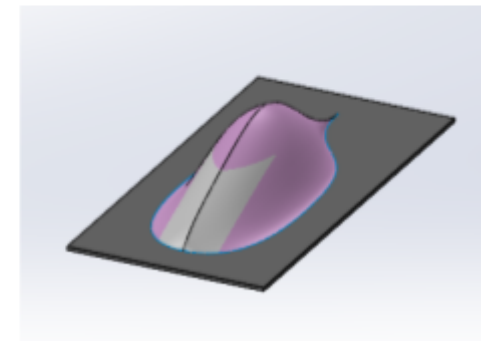
Autonomous Cookie Icing Machine



Mechanical Thumb



Dynamometer



Solar Car Canopy

Autonomous Cookie Icing Machine

Skills: SolidWorks, C++, Fabrication, DFM, Circuits, Machining Safety, 3D Printing

Objective: For the ME101 final design project, design, build, and program an autonomous cookie icing machine capable of applying a user designed multi-color icing pattern to the surface of a cookie in <4 mins.

Outcome: Worked on a four person team to design and build a dual-axis autonomous cookie icing machine consisting of VEX IQ electronics, a gantry system and tri-color rotating syringe barrel, driven by a C++ program. We also created an HTML interface for users to draw custom designs to send to the machine. The machine met all engineering specifications including completing a decoration cycle in <240 secs, accommodate cookies up to 4" in diameter and dispense 3 icing colors.

Experience: I contributed to the 3D modelling of the gantry and icing dispenser in SolidWorks and was responsible for majority of the fabrication, including 3D printing components, such as the motor casings and gantry system, and laser cutting the acrylic racks, baseplate, and enclosure panels. Our initial idea was to have a design where the nozzle moves and the baseplate remains still in the x-y position. Due to having a limited amount of motors, I proposed the idea of moving the baseplate that holds the cookie instead of the icing dispenser to improve icing accuracy and reduce the load on the upper motors. By conducting iterative testing, I also discovered and implemented the logic on the motor-driven plunger to avoid failed dispensing caused by trapped air in the syringe and icing pooling after drawing a figure. Throughout, I applied DFM principles to optimize parts to be low-cost and manufactured quickly while remaining strong.

Learning: I learned the importance of proper calibration of sensors for reliable autonomous operation since they're sensitive and often don't behave exactly as stated on the data sheet.

- ❖ Watch demonstration: [Demo Video](#)
- ❖ Read mechanical design report: [Report](#)

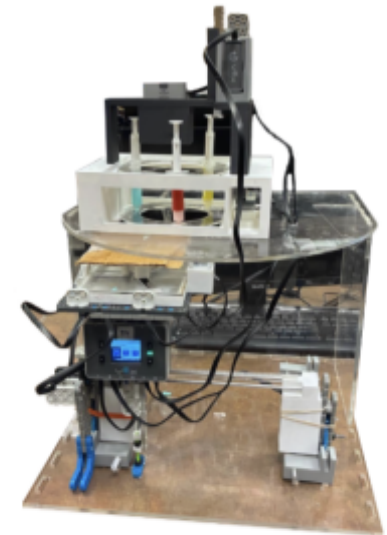
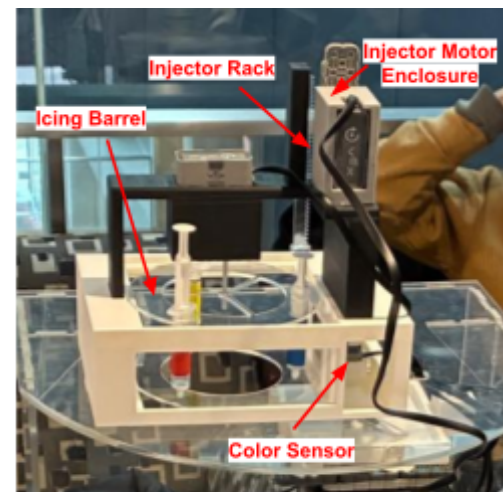
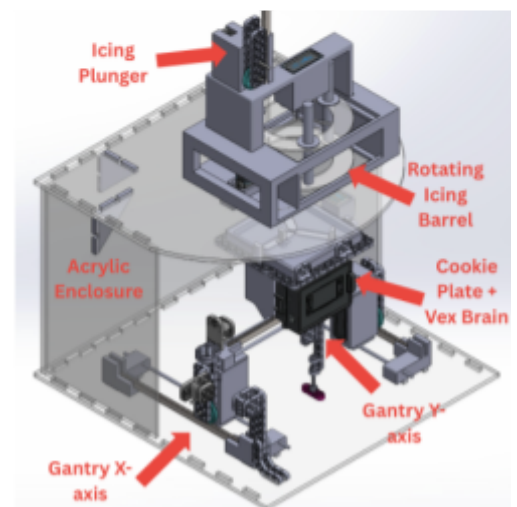


Photo of Design



Icing Dispenser



CAD of Overall Layout

Mechanical Thumb Toy (Digit)

Skills: SolidWorks, Iterative Design, Technical Report Writing, 3D Printing

Objective: Create a mechanically operated toy for the ME100 course final project.

Outcome: Worked on a team to design and prototype Digit, a wearable mechanical thumb capable of actuation and grasping. The purpose of the project was to create a toy using the engineering design process while applying 3D modelling and machine shop skills learned in class. The actuation of Digit is caused by the user's wrist movement and we met a 30N grip force spec with <1 sec actuation.

Experience: I was responsible for several core mechanical components including designing the mounting mechanism of the toy to the user's hand and an elastic extension mechanism that restores the thumb to its neutral position. I modelled the mount in SolidWorks and created iterations of the components based on functional testing. In my initial design, I proposed the idea of having a component that slid onto the user's thumb and attached to a string that was used to move Digit. During testing, I noticed that this configuration defeated Digit's role as an additional finger since it relied on the user's thumb for actuation. To address this, I transitioned to wrist based actuation to free all fingers for use while enabling controlled movement of Digit which we used for the final iteration.

Learning: I learned how to create independent mechanical motion paths that could be connected to specific hand movements, preventing wrist rotation from unintentionally triggering extension or retraction of Digit, and learned to use tension and varying string lengths to actuate Digit.

- ❖ Watch demonstration: [Demo Video](#)
- ❖ Read report detailing Digit's mechanical design and development: [Report](#)



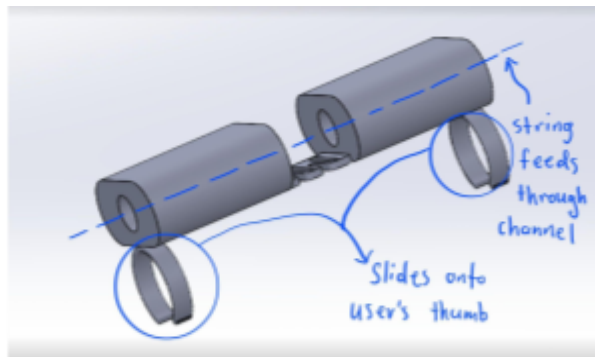
Grasping Object



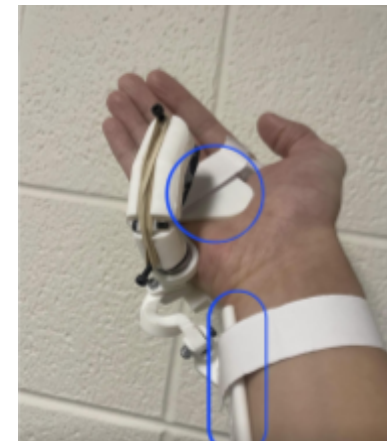
Hand Mount Design



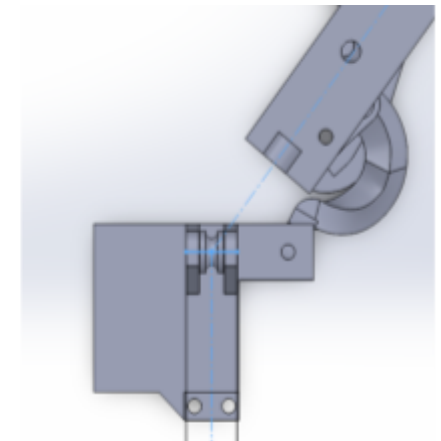
Elastic Extension Mechanism



Thumb Component - 1st Iteration (In SolidWorks)



Hand Mount



Wrist Actuation - Center of Rotation (In SolidWorks)

Solar Car Canopy Design

Skills: SolidWorks, Surface Modelling, Lofting, Split Lines

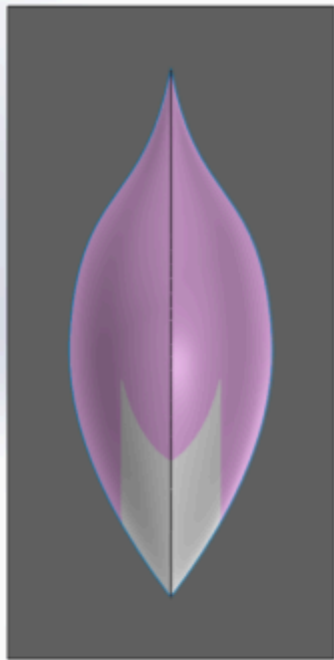
Objective: Design a canopy for the solar car as part of the onboarding process for the University of Waterloo Midnight Sun design team.

Outcome: Created an aerodynamic canopy that underwent approval by the team leads.

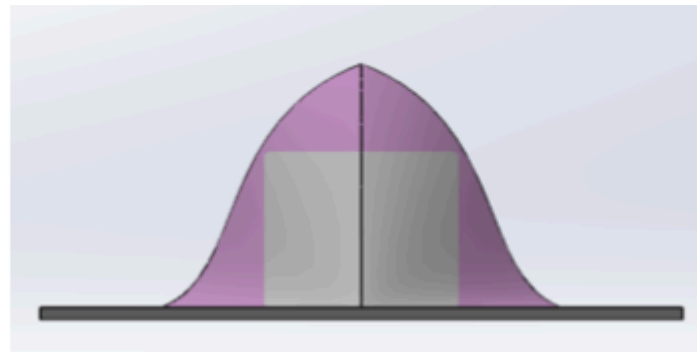
Responsibilities: Designed and modelled a canopy in SolidWorks.

Learning: I learned multiple new SolidWorks modelling techniques including surface modelling, lofting and using split lines.

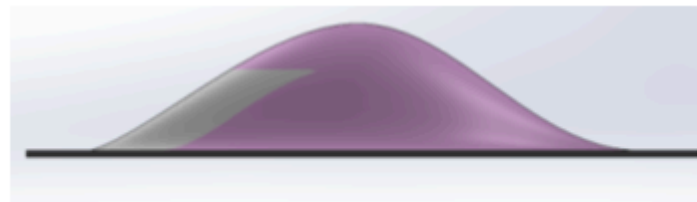
Orthographic views in SolidWorks:



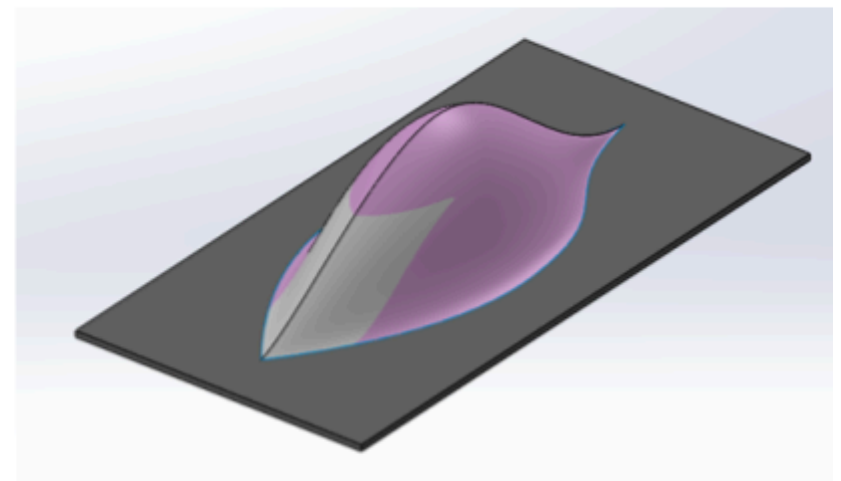
Top



Front



Right Side



Isometric

DIY Dynamometer

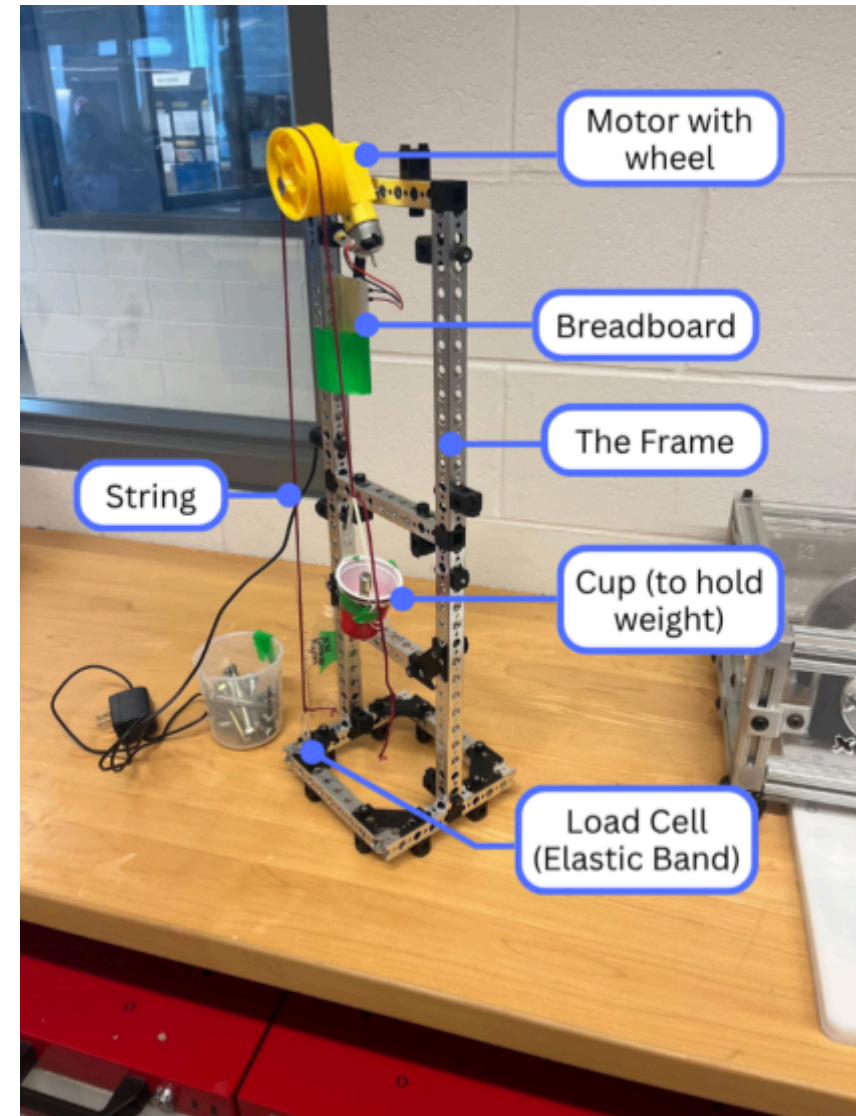
Skills: Excel, Data Analysis, Technical Report Writing

Objective: Design and build a dynamometer to measure torque output of a TT motor using materials from a *TETRIX® Programmable Robotics Set* and a switch, breadboard, jumper wires and power source to operate the motor for a ME100 project. We were also given other materials to optionally use including nuts and bolts (for weights), elastic bands, string and a plastic cup. The project required us to create either a rope or Prony brake dynamometer only using information we found online.

Outcome: I collaborated with four other mechanical engineers to create a rope brake dynamometer since it required less material to build than the Prony brake. We submitted a report evaluating the accuracy of our torque measurements compared to the manufacturer's values and identifying potential sources of inaccuracy.

Responsibilities: I designed the final structure which featured a tall frame that allowed for a wider range of measurements compared to designing the dynamometer to hang off of a table, which was the alternative. I compared the displacement of the elastic band to the elastic force and the torque to the speed of the TT motor in two scatter plots that I created with best fit lines in Excel. I also authored the section of the report about the torque-speed curve and sources of measurement inaccuracy.

Learning: I gained an understanding of the purpose and function of dynamometers, including the difference between rope and Prony brake types, and strengthened my data analysis skills by analyzing correlations in the graphs. I also improved my skills in translating research (about rope brakes) into a physical working product.



Rope Brake Dynamometer

Spur Gear Keychain

Skills: SolidWorks, Machine Shop Equipment Operation

Objective: Design and fabricate a custom keychain.

Outcome: Fabricated a 3.5 x 1.25 x 0.5" keychain with a revolving spur gear.

Experience: I modelled the full assembly in SolidWorks and utilized a lathe, 3-axis mill machine, drill press and a tap (for internal threads) to create and assemble the part.

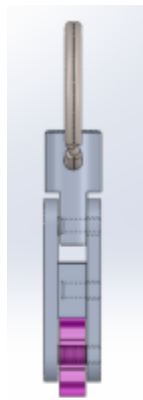
Learning: I learned to execute clean and precise machining and learned about proper machine shop etiquette and safety. I also learned to model physical mechanisms in an assembly in SolidWorks. **See below for orthographic views in SolidWorks:**



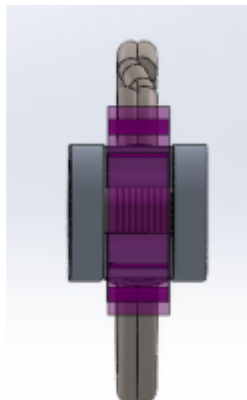
Assembled Keychain View 1



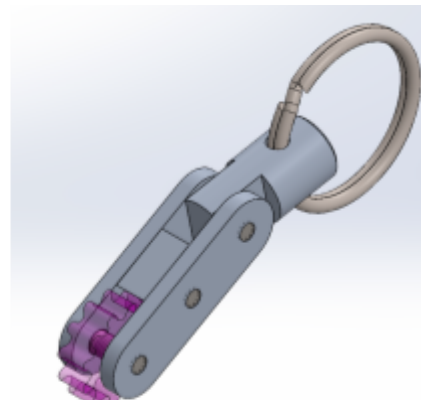
Assembled Keychain View 2



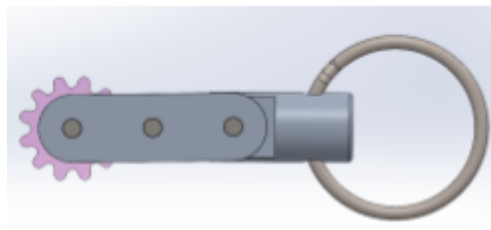
Top



Front



Isometric

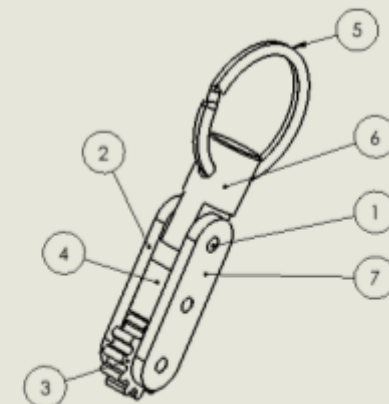


Right Side



Assembled Keychain View 3

ITEM NO.	PART NUMBER	DESCRIPTION	QTY.
1	M4 Screws	AISI 1020	3
2	Top Plate	3003 Alloy Aluminum	1
3	Spur Gear	Acrylic	1
4	Spacer	3003 Alloy Aluminum	1
5	Keyring	Alloy Steel	1
6	Keyring Connector	3003 Alloy Aluminum	1
7	Bottom Plate	3003 Alloy Aluminum	1



Bill of Materials